

Applied SVAR Identification

Jamie L. Cross

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Course details

- **Dates:** Monday 22 June – Friday 26 June 2026
- **Time:** 10:00–12:30 each day (12.5 lecture hours total)
 1. Part 1: 10:00–11:10
 2. 10-minute break: 11:10–11:20
 3. Part 2: 11:20–12:30
- **Format:** Integrated discussion of theory and applications each day
- **Software:** MATLAB – primarily using [Ambrogio Cesa-Bianchi's SVAR Toolbox](#)

Schedule

Day 1 — VAR Foundations

- Motivation: From dynamic regressions to SVARs
- Identification: Reduced form vs Structural VARs
- Estimation: OLS and Bootstrapping
- Specification: Transformations, Lags, and Stability
- VAR Tools: Granger Causality and Forecasting
- SVAR Tools: IRFs, FEVDs and HDs
- *Application:* Stock and Watson (2001)

Day 2 — Identification I: Short-run Zero Restrictions

- Short-run restrictions: Recursive identification
- *Applications:* Kilian (2009); Wong (2015); Blanchard and Quah (1989); Bjornland (2009)

Day 3 — Identification II: Long-run Zero Restrictions

- Long-run restrictions: Blanchard–Quah identification
- *Applications:* Kilian (2009); Wong (2015); Blanchard and Quah (1989); Bjornland (2009)

Day 4 — Identification III: Sign, Magnitude and Narrative Restrictions

- Sign restrictions
- Magnitude restrictions
- Narrative restrictions
- *Applications*: Uhlig (2005); Kilian and Murphy (2012); Antolin-Diaz and Rubio-Ramirez (2018)

Day 5 — Identification IV: Instruments

- Proxy-SVARs
- VARX models
- *Applications*: Gertler and Karadi (2015); Känzig (2021); Paul (2020)

References

- [1] Antolín-Díaz, J., and J.F. Rubio-Ramírez (2018). Narrative sign restrictions for SVARs. *American Economic Review*, 108(10), 2802–2829.
- [2] Bjørnland, H.C. (2009). Monetary policy and exchange rate overshooting: Dornbusch was right after all. *Journal of International Economics*, 79(1), 64–77.
- [3] Blanchard, O.J., and D. Quah (1989). The dynamic effects of aggregate demand and supply disturbances. *American Economic Review*, 79(4), 655–673.
- [4] Gertler, M., and P. Karadi (2015). Monetary policy surprises, credit costs, and economic activity. *American Economic Journal: Macroeconomics*, 7(1), 44–76.
- [5] Känzig, D.R. (2021). The macroeconomic effects of oil supply news: Evidence from OPEC announcements. *American Economic Review*, 111(4), 1092–1125.
- [6] Kilian, L. (2009). Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American Economic Review*, 99(3), 1053–1069.
- [7] Kilian, L., and D.P. Murphy (2012). Why agnostic sign restrictions are not enough: Understanding the dynamics of oil market VAR models. *Journal of the European Economic Association*, 10(5), 1166–1188.
- [8] Paul, P. (2020). The time-varying effect of monetary policy on asset prices. *Review of Economics and Statistics*, 102(4), 690–704.
- [9] Stock, J.H., and M.W. Watson (2001). Vector Autoregressions. *Journal of Economic Perspectives*, 15(4), 101–115.
- [10] Uhlig, H. (2005). What are the effects of monetary policy on output? Results from an agnostic identification procedure. *Journal of Monetary Economics*, 52(2), 381–419.
- [11] Wong, B. (2015). Do inflation expectations propagate the inflationary impact of real oil price shocks? Evidence from the Michigan Survey. *Journal of Money, Credit and Banking*, 47(7), 1673–1693.